



Hobbies:

- Watching movies
- Playing soccer

Favorite things to do:

- Traveling
- Hiking
- Watching soccer games

Mohamed Elkasaby “Sherif”
I was born in Egypt and came to the United States in 2016.
I currently live in Rock Hill, SC.

Transferred to

York Tech
(Aug 2019- Dec 2020)

Clemson University
(Jan 2021- Present)

Major
Mechanical engineering



First Co-op rotation
with GE Aerospace
(Jan 2023-May 2023)

Worked With
Hybrid Electric team at
Evendale, Cincinnati, Ohio



Second Co-op rotation
with GE Aerospace
(Sep 2023-Dec 2023)

Worked With
Aeromechanic team
at Evendale, Cincinnati,
Ohio

Clemson University
(Present –August 2024)

Preparing to graduate
in **August!**



Major
General engineering



Go Tigers!

“Fun fact about me,
I am addicted to Coffee .”



Exciting Time

- I participated in two Co-Op rotations with GE Aerospace. What leads me back to GE is the **diverse culture and amazing experiences I had with the different teams I was a part of, along with the new technical competencies gained. I have since, leveraged those competencies into my Capstone project.**
- I will be successful in this program because I am passionate when it comes to the applied sciences. The technical roles offered within the Edison Program inspire initiative and curiosity in me that fuels my drive to further my technical prowess.
- My career aspiration is to get involved in **Structural analysis (Vibration).**

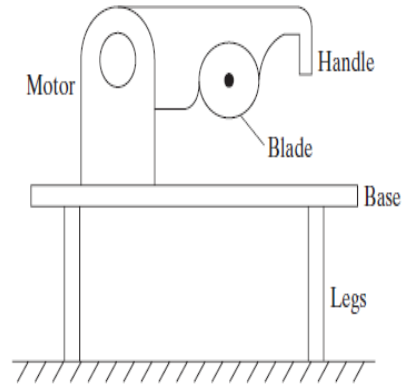
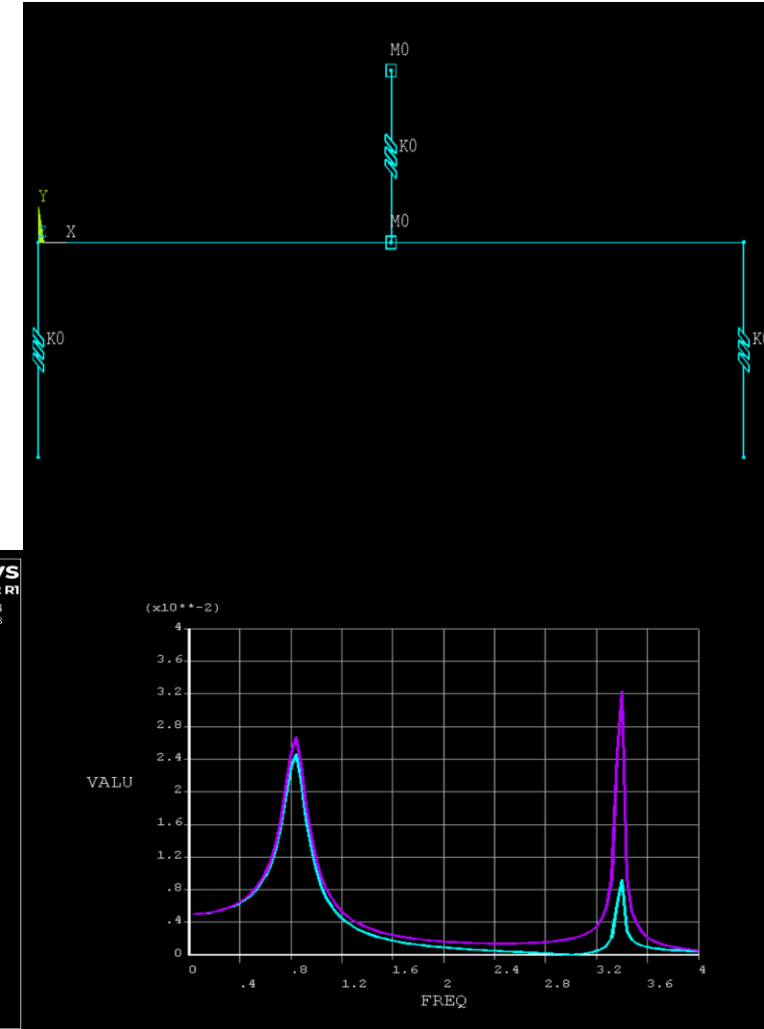
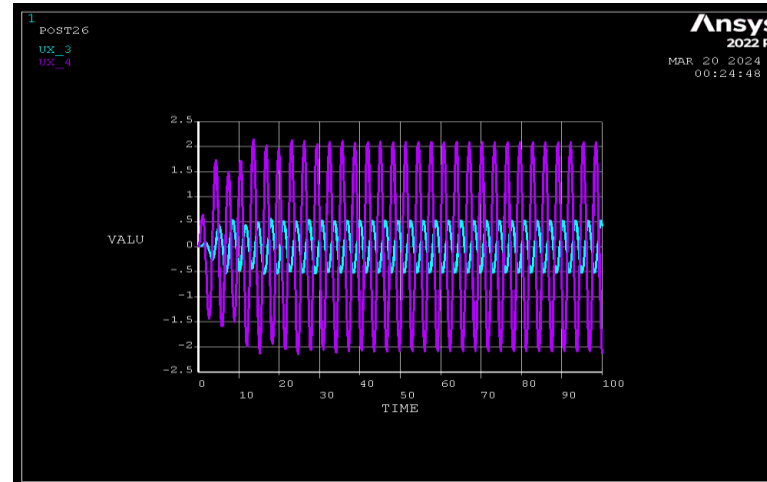
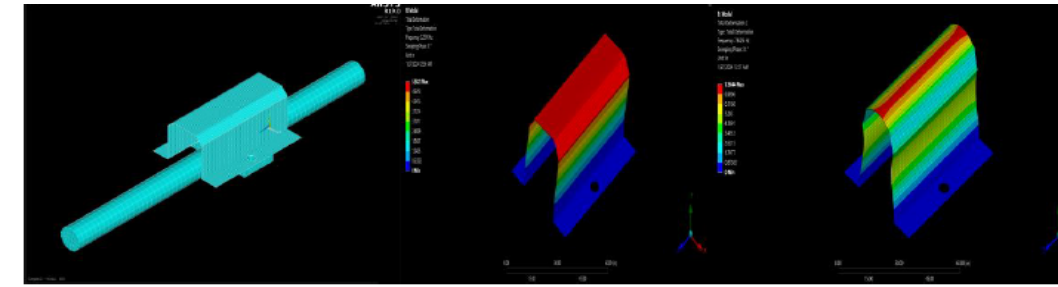
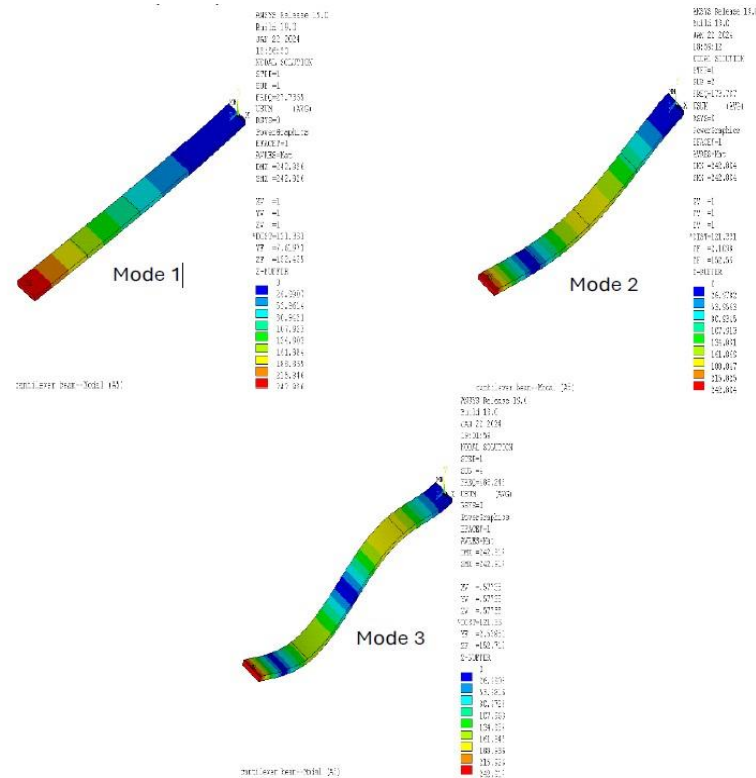
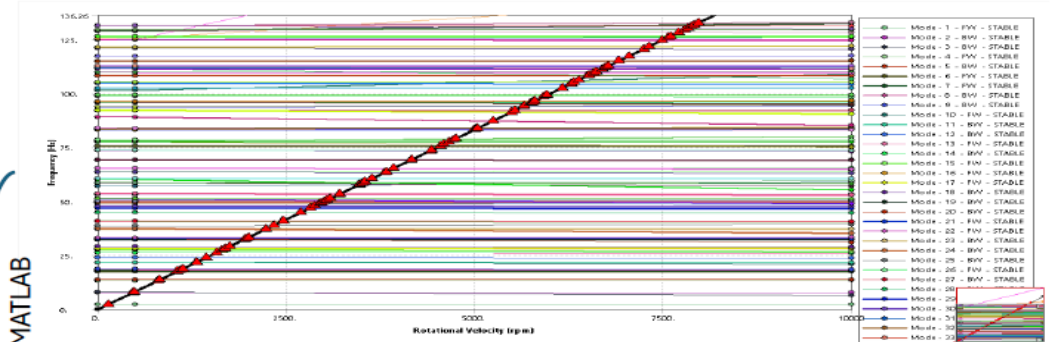


Figure 5.17 A schematic of a radial saw system in need of a vibration absorber.



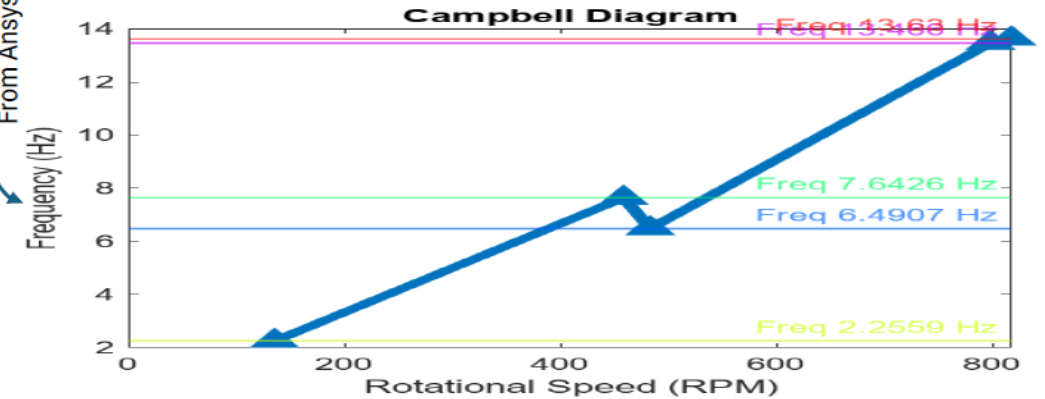


Model representation First Mode shape Second Mode shape



Campbell diagram from Ansys

From Ansys to MATLAB



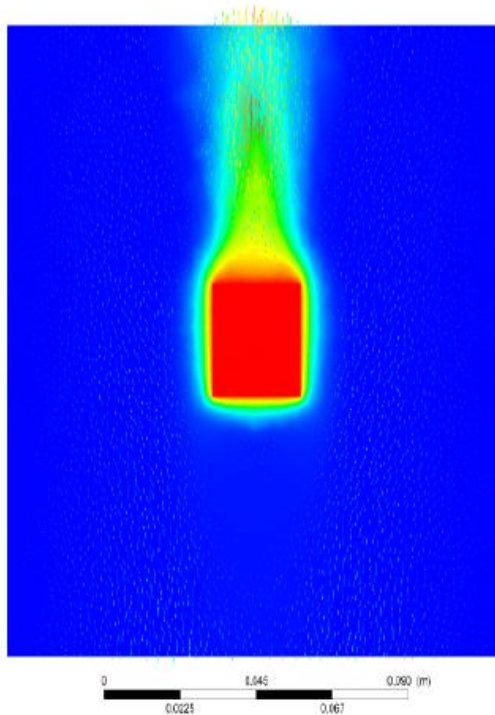
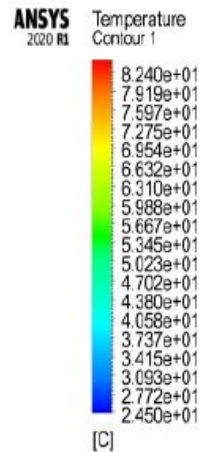
Campbell diagram for first 4 modes in MATLAB

Natural frequency (HZ)	First Mode (Hz)	Second Mode (Hz)	Third Mode (Hz)
Theoretical Results	27.596	172.97	484.358
Numerical Results (FEM)	27.73	173.707	486.248
Experimental Results	25.547	172.517	---

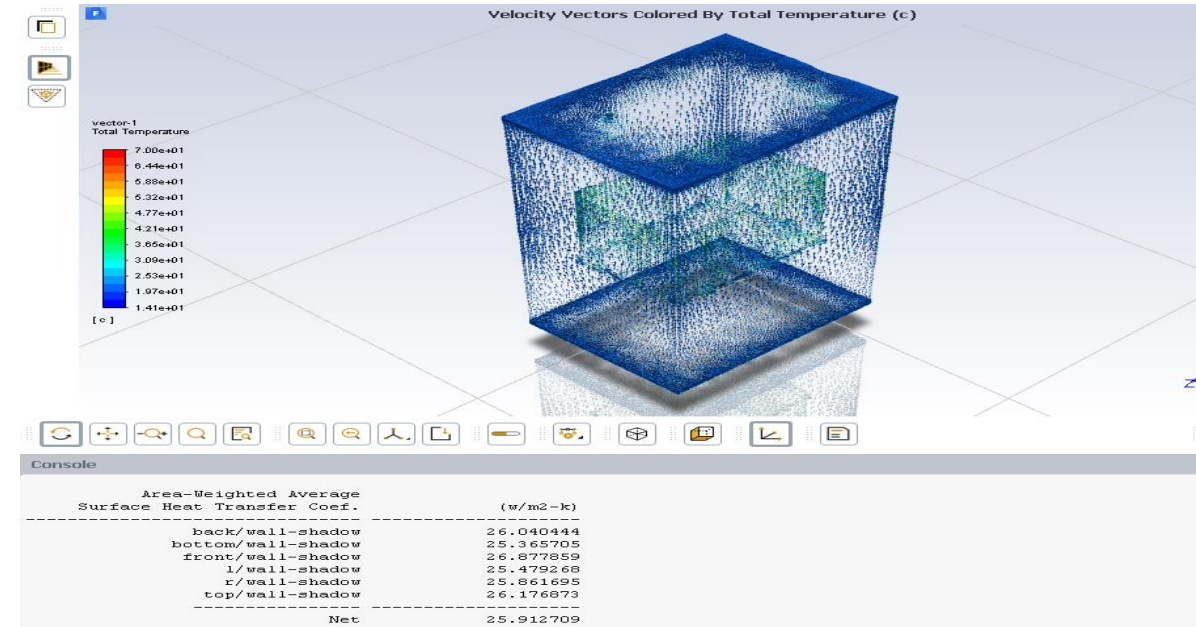
Finite element analysis (FEA)

Technical experience

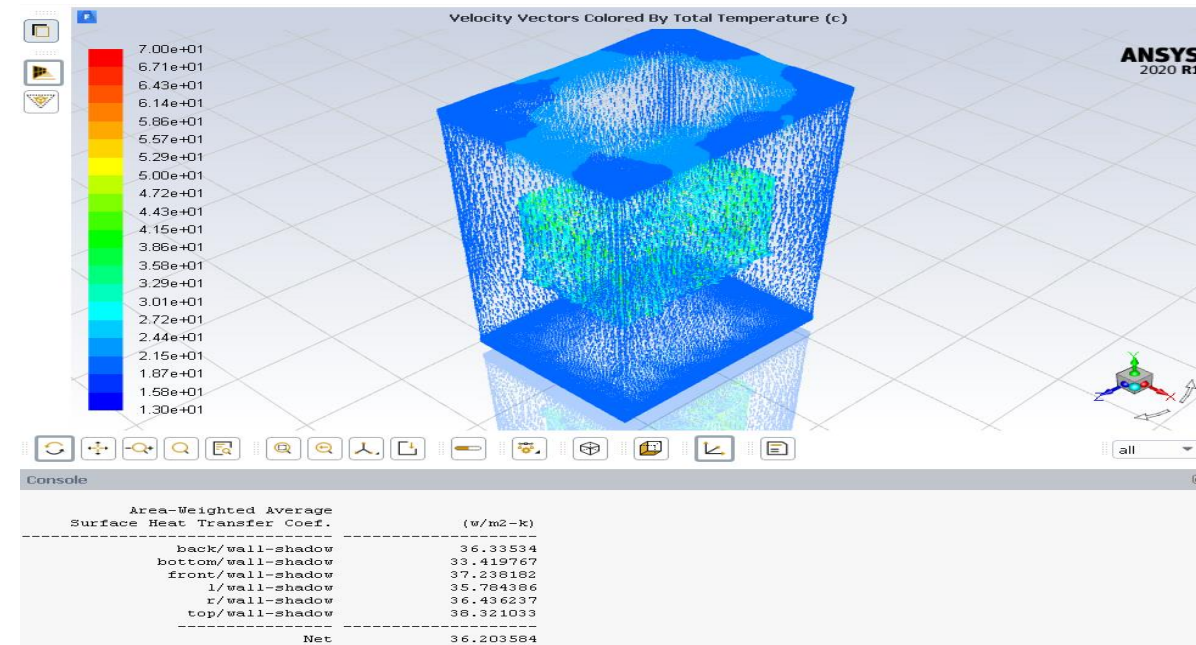
Computational fluid dynamic (CFD) : Ansys Fluent



Natural Convection simulation

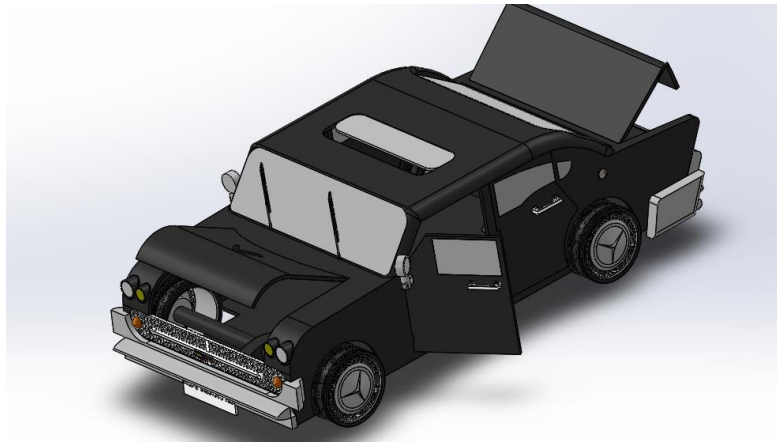


Forced Convection simulation at different air speed



Technical experience

CAD 3-D modeling:

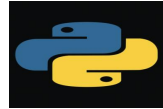
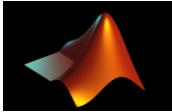


SolidWorks

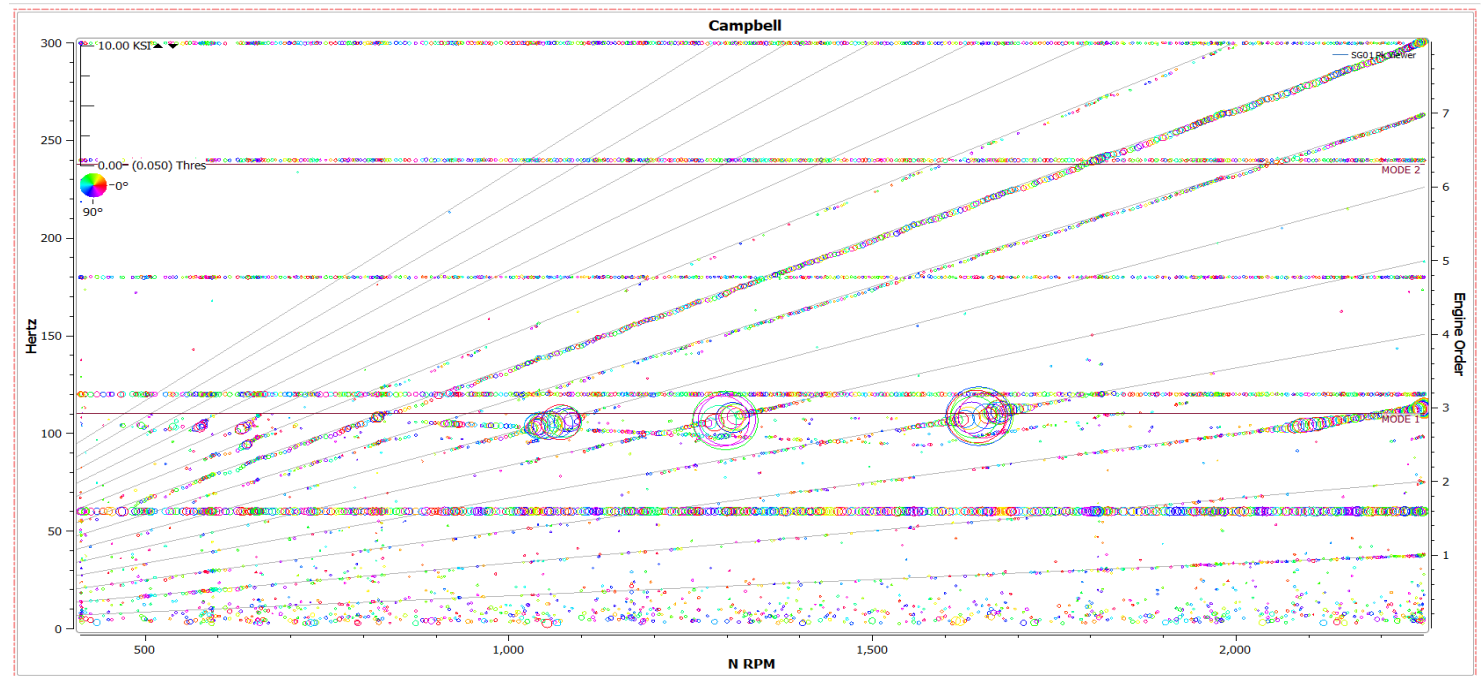


NX siemens

Programming language:



Data Processing Software:



Campbell Diagram

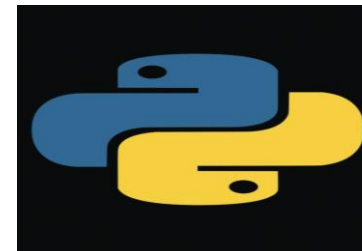
Team experience

My Role: GE intern with Hybrid Electric team

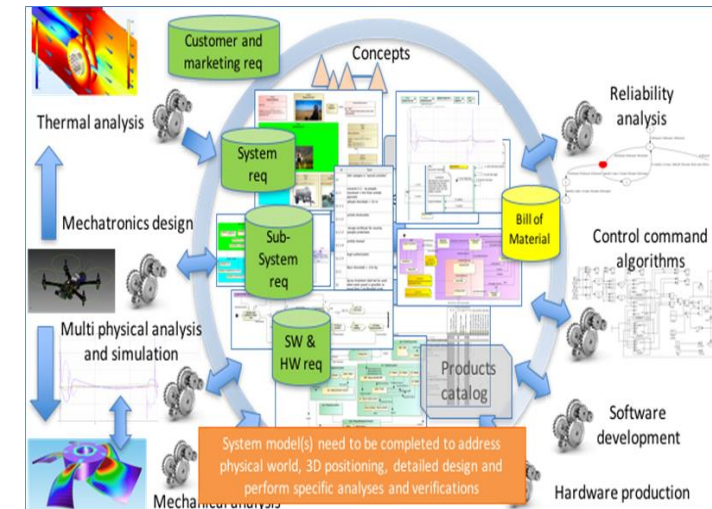
- Conducted investigations into previous tests (they were done on hybrid components of the engine)/ Prepared a detailed technical presentation.
- Engineered a robust plotting tool using Python. This tool significantly improved the visualization of the engine test data, making complex information more accessible and understandable for tests were done on hybrid components of GE ct7-9B engine.
- Authored comprehensive documentation that serves as a user-friendly guide for the plotting tool. This ensures that others can easily utilize and benefit from the tool's capabilities.
- Standardized work processes for data handoffs with global. The goal is to share information with global teams efficiently. I played a key role in GE teams By leveraging Cameo, A model-based engineering system software, I created efficient and reliable procedures that streamline collaboration across different teams and time zones.



Hybrid Electric Team at GE Aerospace



Plotting tool



Model-based systems engineering

Team experience

My Role: GE intern with Aeromechanic team

- Performed modal analysis on enhanced airfoil geometries using Ansys APDL. This helped to provide a feedback and improving the new tool to leverage for Aeromechanic team.
- Developed a Python script to automate the creation of a parameter database for Apex DX. By utilizing historical data, this script streamlined the process, saving significant time and reducing potential errors.
- Formalized and standardized, documentation of the creation process of the Apex DX parameter database and provided detailed instructions for seamless operation of the automated tool.



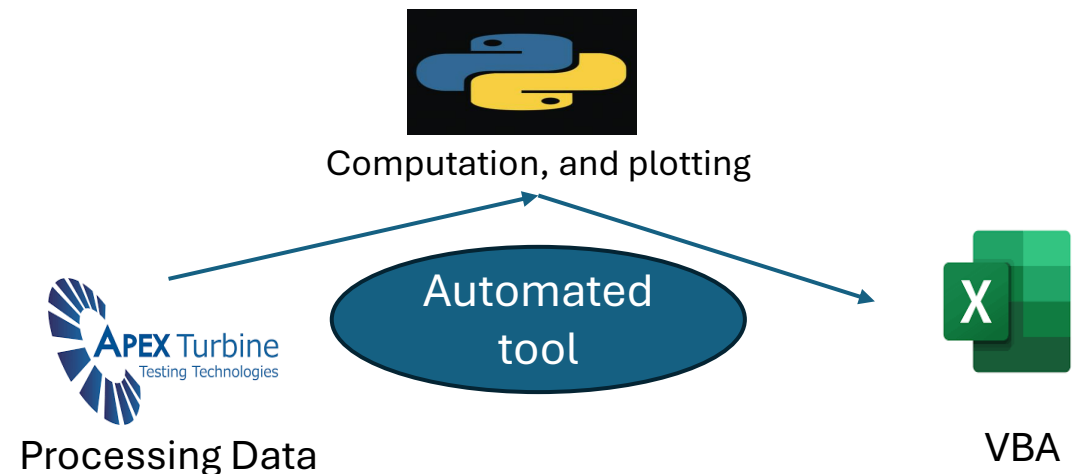
Model Analysis



e-Blade



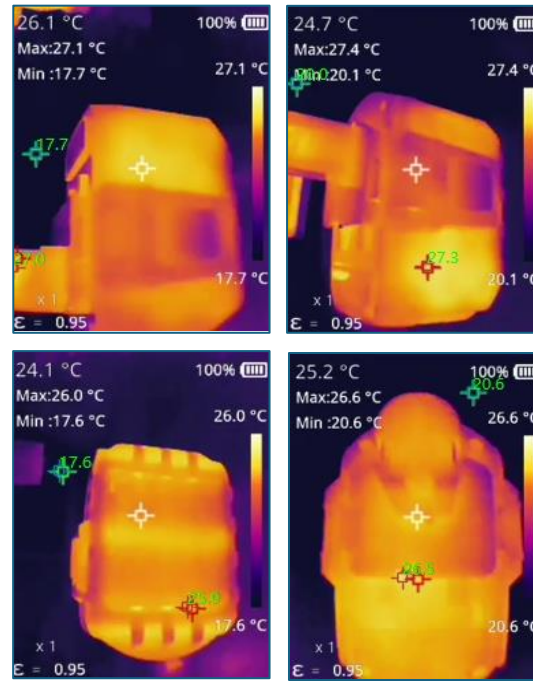
Aeromechanic Team at GE Aerospace



Team experience

My Role: Test Engineer

- Developed methods to measure the temperature across each battery surface to identify high temperature concentrations.
- Modeled and simulated various design concepts for optimal selection.
- Conducted CFD analysis to evaluate the performance of the final design.
- Applied FEA analysis for validation, following DFMEA guidelines.
- Implemented additional methods to enhance the design and provided recommendations for further improvement.



Background:

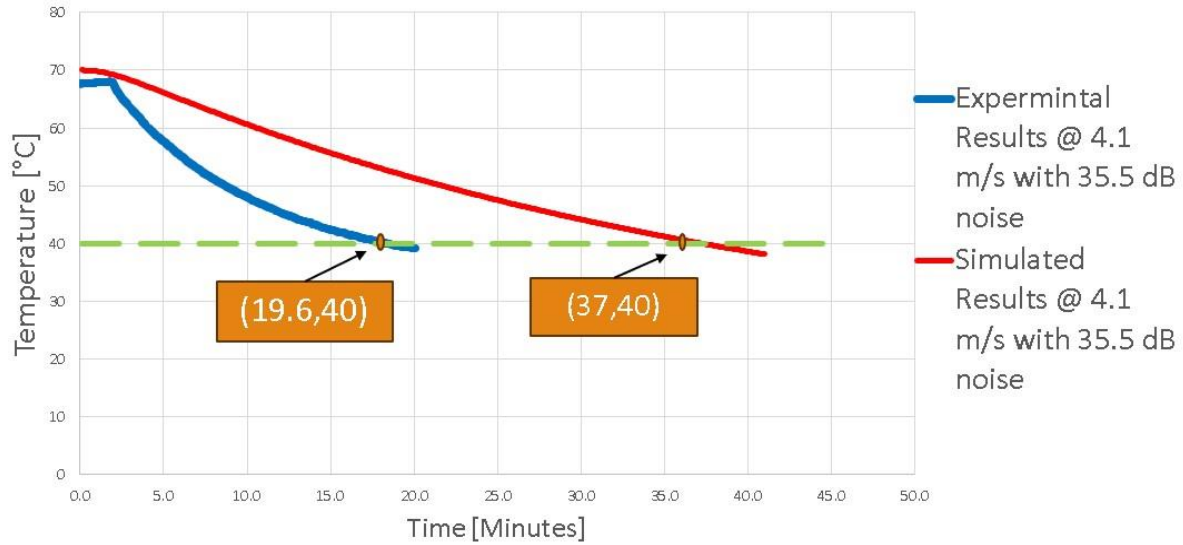
- Ryobi is currently addressing the wait time for a battery to initiate a charge and the charge time in hot conditions
- Batteries tend to shutdown due to overheating and charging of the battery is prevented when the battery is over a specified temperature

Goal:

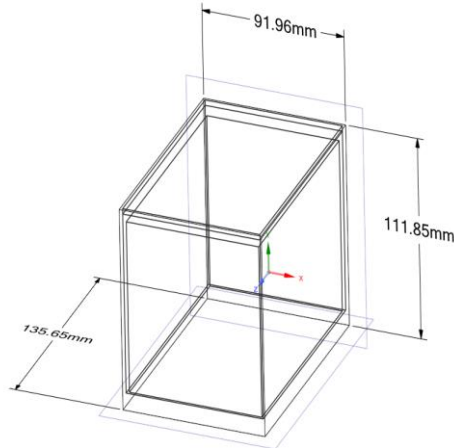
- Develop a cost-effective method to cool battery cells when they are on a charger

The main requirements of focus are:

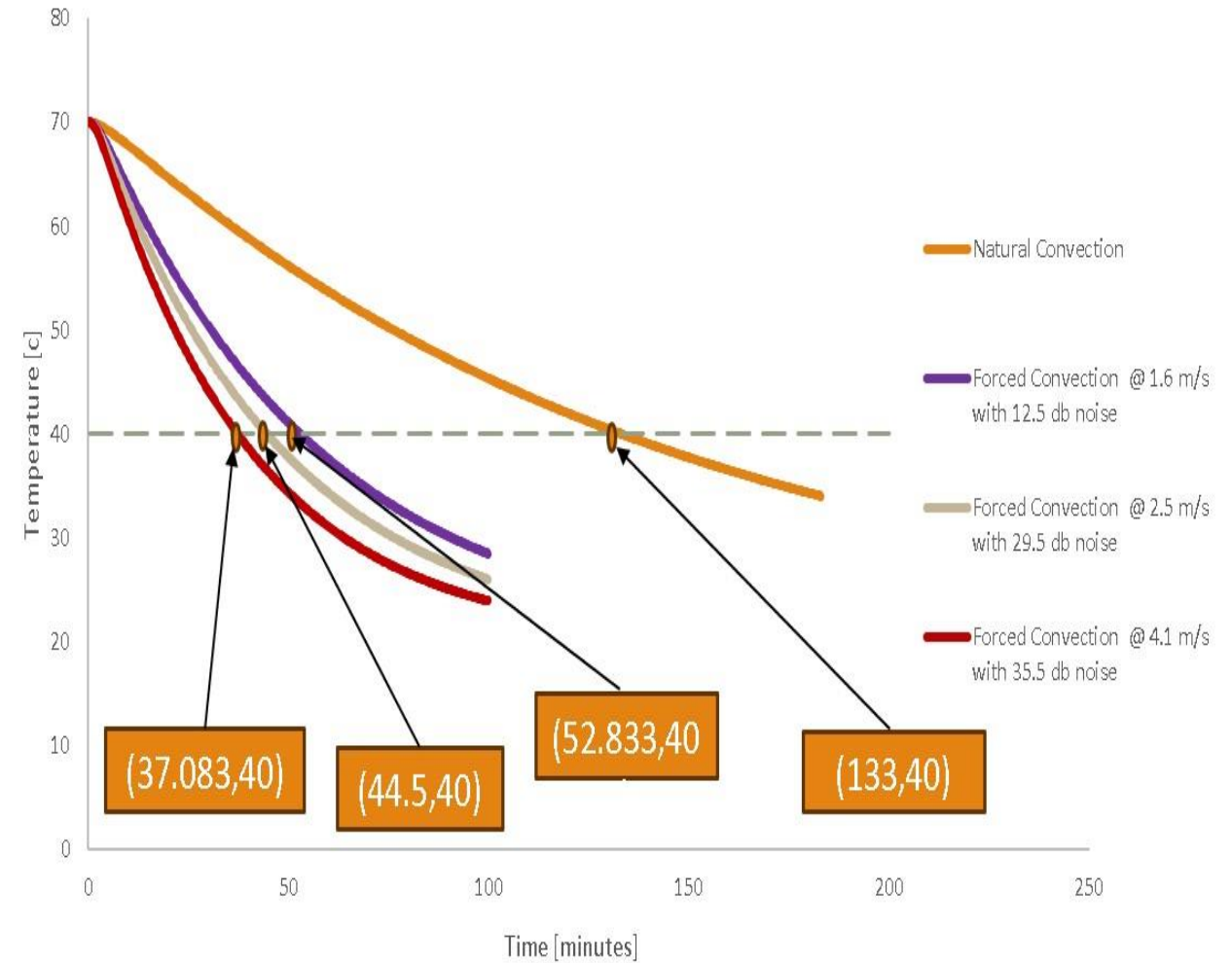
- The wait time for the battery to receive charge from overheating must be no longer than 31.5 minutes.
(The system must reduce the wait time after overheating by 30%)
-
- The charge time for a fully depleted battery in a 40°C environment must be no longer than 76 minutes.
(The system must reduce the charge time in a 40°C chamber by 20%)



Comparison with simulated results and Experimental results

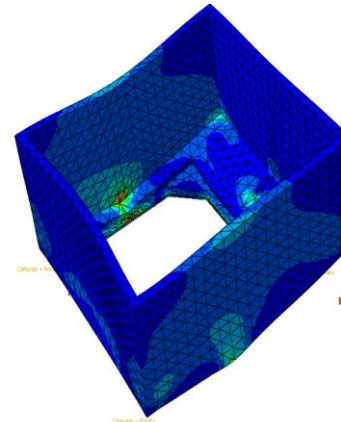
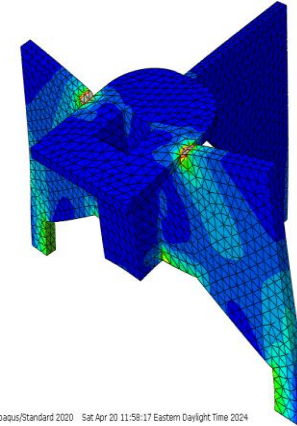
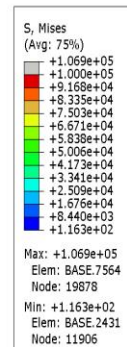
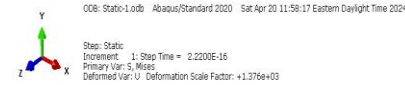
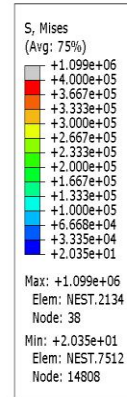
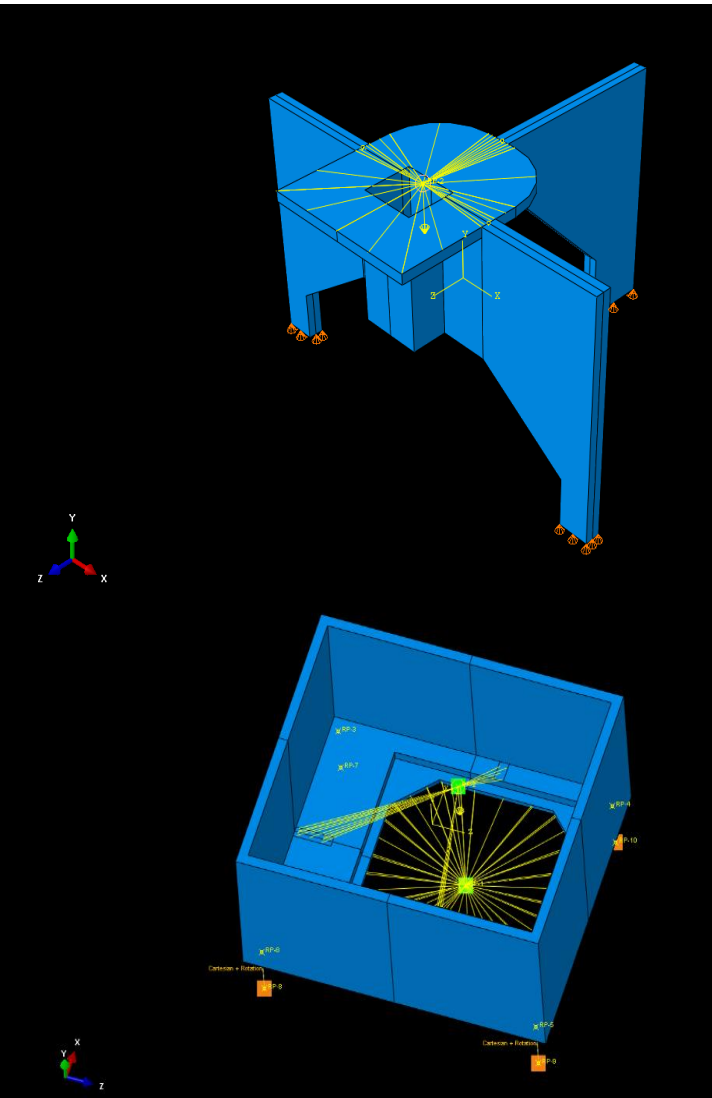


Model Consist of ABS Material, Air, and Lithium

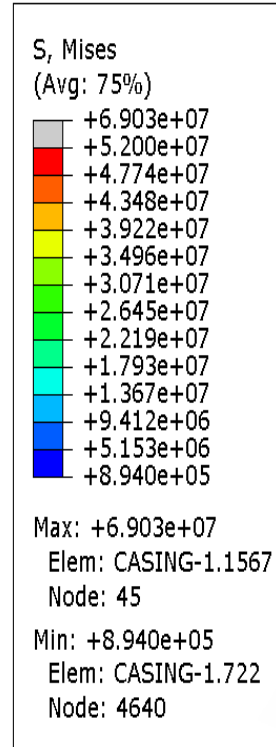


Simulated results at different speed

Team experience



Static analysis



ODB: Job-2.odb Abaqus/Explicit 2020 Sat Apr 20 13:38:13 Eastern Daylight Time 2024

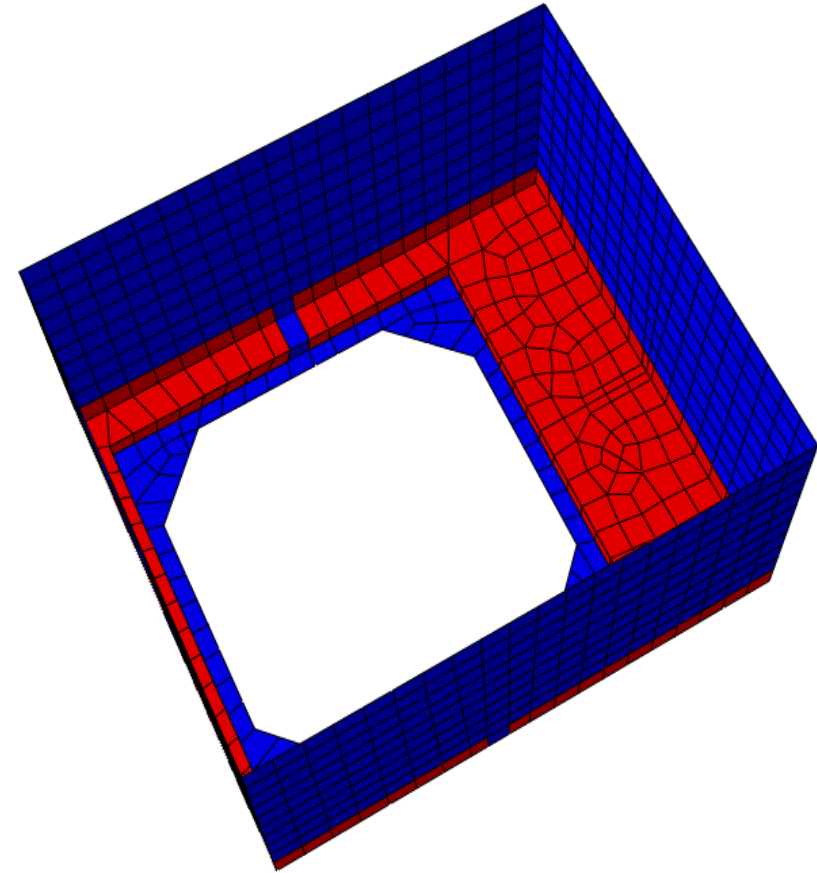
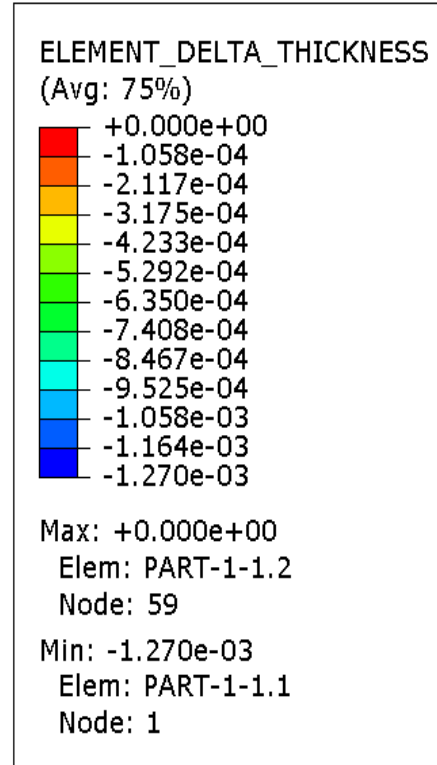
Step: Step-1
Increment: 1343 Step Time = 5.0000E-04
Primary Var: S, Mises
Deformed Var: not set Deformation Scale Factor: not set

Shock analysis

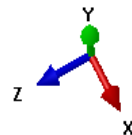


Optimization

Decreasing the Thickness from 6.35 mm to 5.08 only blue region of the casing.

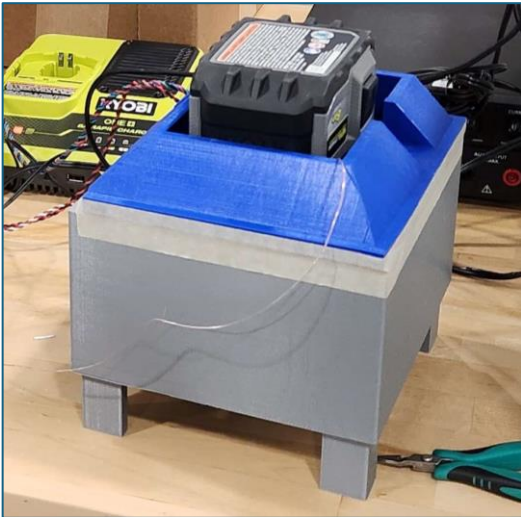
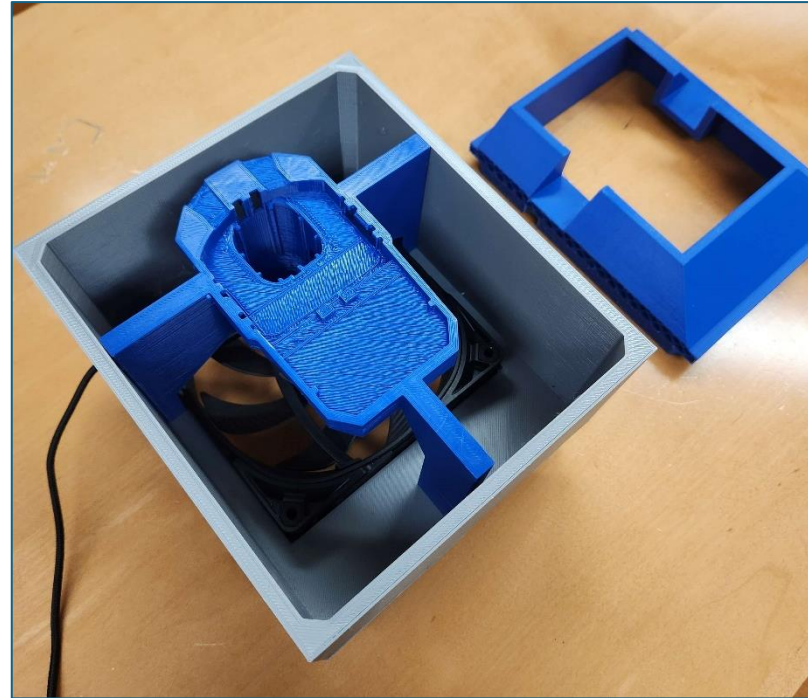


ODB: Sizing-Job_post.odb Abaqus/Standard 2020 Sat Apr 20 14:33:30 Eastern Daylight Time 2024



Step: Static_Optimization, Step containing results of optimization design cycles for analysis step: Static, baseODB=4
Optimization Design Cycle-4
Primary Var: ELEMENT_DELTA_THICKNESS

'Bottom Fan' Prototype



Vibration (MODF)

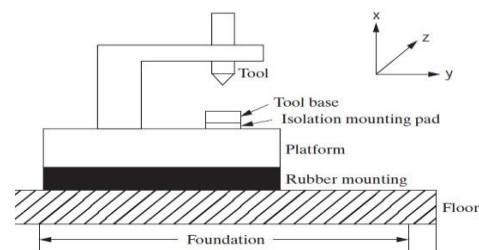


Figure 4.27 A schematic of a punch-press machine.

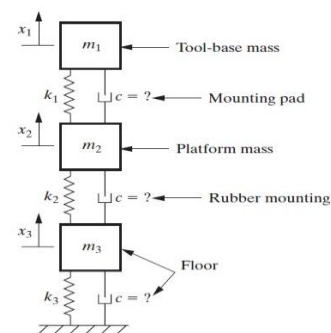
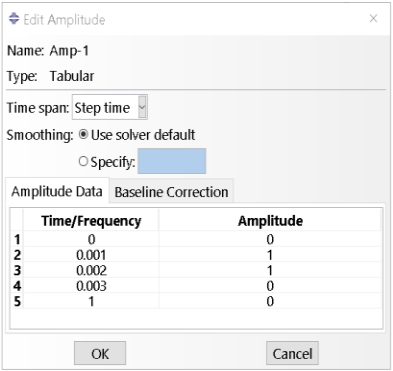
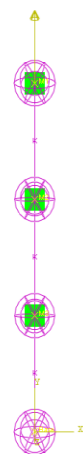
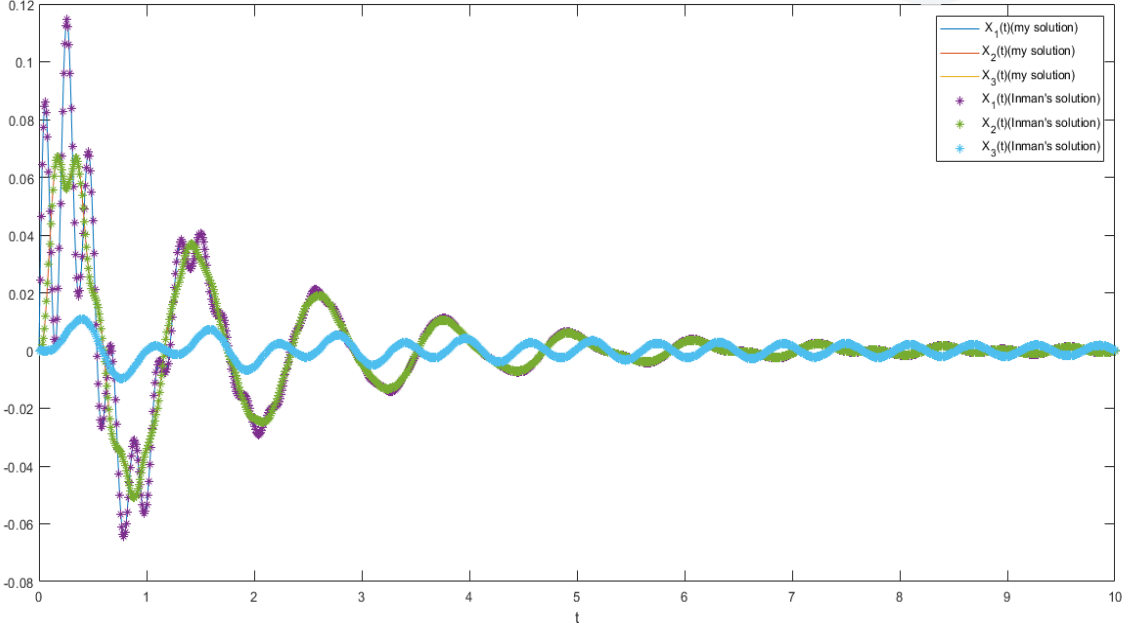


Figure 4.28 A vibration model of the punch press of Figure 4.27.



ModalSuperposition Method in MatLAB



FEA

